

Board notes

Print this and carry it. The deck records what is *projected*; this records what gets *written*.

This session is not a board session. One derivation, on the split surface. The board carries the **carried-over toggle lines** from Thursday, the **delay convention**, and the **group answers** from the argument at 14–24.

Conventions. **ASK** — put it to the room before writing, and the answer to expect. **POINT** — the moments you turn and point at the board rather than the screen. **IF ASKED** — the question that always comes.

Before class — left wing, the two lines Thursday did not reach

Session 9 ran out at its slide 20, so the failure surface and the parameter window are carried into the opening. Write these two before class:

the wedge is finite

failure = walking out of it

Leave room underneath. The four failure mechanisms come from the room in the first five minutes and get written there:

mutation in a repressor • promoter mutation • plasmid loss • burden selecting against the expressing state

ASK before writing any of them: *a mutation that breaks the circuit, and selection that breaks the population — are those the same problem?* They are not, and they need different fixes: sequence redundancy against the first, lower burden against the second. That distinction is the one to get on the board, not the list.

Centre — the delay convention, written once, stays up all period

This is the T25 specification and it is on the handout. Write it in full, and **do not erase it** — the faded set at 63 refers back to it and PS5 marks it.

$t_{1/2}$: **Z reaches 50% of its steady state**

$t_D = t_{1/2}(\text{FFL}) - t_{1/2}(\text{simple, same steady state})$

POINT at it twice: once when the derivation reaches step 3, and again at 63 when you send them to the handout.

ASK when you write it: *why does the comparison circuit have to reach the same steady state?* Because otherwise you are timing two different journeys — half of a big number and half of a small one are not comparable, and the difference would be an artifact of the destination rather than the route.

Right wing — minutes 14–24, the group answers

Three columns, one per question, answers **by group name**. You read them back at 24 and sort them into the three rows of the next surface. Do not sort while writing.

Expect column 1 to be easy (the CI arm is slow) and column 3 to be the hard one. If no group gets column 3, do not supply it — the sorted slide's third row is where it lands, and the figure does the arguing.

ASK at 24, before sorting: *did anyone's answer to 3 use the word rate rather than the word amount?* That is the distinction the whole result turns on, and naming the group that said it first is worth more than saying it yourself.

During the derivation — the two lines that go on the board

The run is on the split surface, so the board takes only what has to survive it.

At step 3, write:

$$t_D = \alpha_y^{-1} \ln \left[\frac{Y_{\max} - Y_{\min}}{Y_{\max} - K_{yz}} \right] = \ln 2 \approx 0.7 \text{ at } K_{yz} = \frac{1}{2}$$

CHECK out loud: put $K_{yz} \rightarrow Y_{\max}$ in your head — the log blows up, so the delay is unbounded as the threshold approaches Y 's ceiling. Say that it is *the formula* that blows up, and that the handout asks whether the circuit does. It does not, and that is item 4.

At step 4, write:

$$\text{OFF step : } t_D = 0$$

POINT at the two together. The asymmetry is the session's result and it should be visible in one glance.

IF ASKED — "is the formula exact?" No, and the handout makes that the question. It is the sharp-gate limit; at $H = 2$ it is wrong in both directions, and raising H drives the simulation onto it. Do not give the numbers away — they are on the handout and on PS5.

If a group reaches type 4 on item 4 — and someone will

They are right, and it is the better answer. Do not wave it away and do not work it at the front; send them to the answer sheet afterwards. What to say, in one sentence, and it is worth writing on the board:

type 1: Y CLIMBS to K_{yz} (bounded by $Y_{\max}$) type 4: Y FALLS past K_{yz} (unbounded)

POINT at the two and say: a timer built on an approach to a ceiling has a ceiling; a timer built on a decay does not. Type 4 hits the hour at $H = 2$ with $K_{yz} \approx 0.17$, no cooperativity needed. That is the item.

Ledger — add to the running list, right-hand column

t_D and its convention · adaptation error = $Z_{\text{final}}/Z_{\text{peak}}$
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Two entries. The adaptation error goes up at 45, when the number surface is on the screen, and stays for the handout.

Closing — the one line to leave up

coherent $\Rightarrow$ delay incoherent $\Rightarrow$ pulse, adapt, accelerate all of it, one direction only

Leave it. Thursday's retrieval opens against it.